Corn leaf diseases: Gray leaf spot vs. Northern Leaf Blight

Oct 26, 2026 · @Marcell

Question: is it plausible that the corn leaves our model confuses carry both diseases at once? **Answer: yes, for several of them, but not all.** Five of the 14 test images (#2, #5, #7, #12, #13) show an NLB-like lesion plus a separate GLS-like area, and 54% of NLB-labelled validation leaves show the same pattern. One error (#8) looks like GLS only, which fits a mislabel better.

Results

A 21-class EfficientNetB0 trained on the PlantVillage color images reaches 98.8% validation accuracy, but gets only 9 of the 14 held-out test images right. Every error is a Northern Leaf Blight (NLB, *Exserohilum turcicum*) leaf called Gray leaf spot (GLS, *Cercospora zeae-maydis*).

- **Data:** 54,305 images, labelled by disease only (21 classes). The 14 test images from `Guess.txt` were excluded; 10% of the rest was held out for validation (5,430 images).
- **Weakest pair:** GLS (F1 0.84) and NLB (F1 0.90) are the two lowest classes. Most other classes score 0.94–1.00, including smaller ones with about 270 images.
- **Not a data-volume problem:** GLS has 513 images and NLB 985, more than several classes the model handles almost perfectly.
- **The model sees a two-way choice:** on every test image the top two guesses are GLS and NLB; the third is under 1% in all but one case.
- **Small test set:** with 14 images, 9/14 fits a true accuracy anywhere from about 40% to 85%.

Predictions on the 14 test images

#	Folder label	1st guess	2nd guess	3rd guess	Correct?
1	NLB	NLB 88.3%	GLS 11.6%	Apple scab 0.0%	yes
2	NLB	GLS 57.5%	NLB 42.1%	healthy 0.3%	**no**
3	NLB	NLB 85.0%	GLS 14.7%	Bacterial spot 0.2%	yes
4	GLS	GLS 98.1%	NLB 1.8%	Isariopsis leaf spot 0.0%	yes

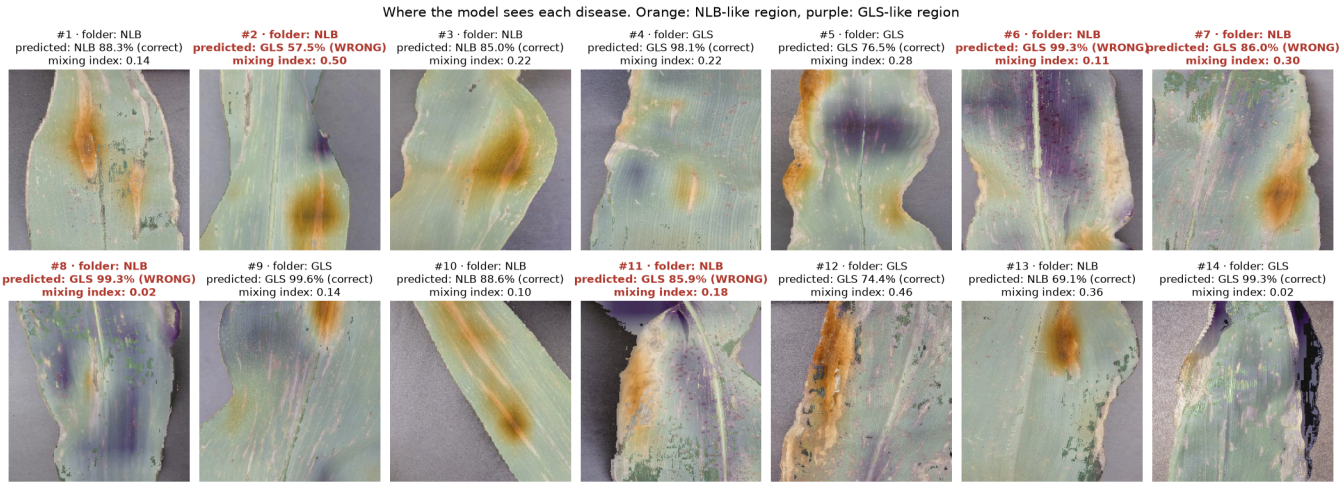
#	Folder label	1st guess	2nd guess	3rd guess	Correct?
5	GLS	GLS 76.5%	NLB 11.7%	Bacterial spot 11.5%	yes
6	NLB	GLS 99.3%	NLB 0.7%	Isariopsis leaf spot 0.0%	**no**
7	NLB	GLS 86.0%	NLB 13.9%	Isariopsis leaf spot 0.0%	**no**
8	NLB	GLS 99.3%	NLB 0.7%	Powdery mildew 0.0%	**no**
9	GLS	GLS 99.6%	NLB 0.4%	Common rust 0.0%	yes
10	NLB	NLB 88.6%	GLS 10.8%	healthy 0.3%	yes
11	NLB	GLS 85.9%	NLB 13.5%	Common rust 0.5%	**no**
12	GLS	GLS 74.4%	NLB 25.5%	Leaf scorch 0.0%	yes
13	NLB	NLB 69.1%	GLS 30.8%	Bacterial spot 0.0%	yes
14	GLS	GLS 99.3%	Apple scab 0.6%	Common rust 0.1%	yes

Confidences are averaged over the original image and three flipped copies. The folder label is the dataset's label and was used only for scoring, never shown to the model.

Why the model confuses them

Many misclassified leaves show two separate regions, one NLB-like and one GLS-like, which points to co-infection that single-label PlantVillage cannot represent.

The model's last step averages an 8×8 feature grid and then scores it. That lets us split each disease score exactly by leaf region (a class activation map). From the per-region difference between the NLB and GLS scores we compute a **mixing index**: 0 when one disease dominates the whole leaf, 0.5 when both are equally present in different places.



Above each leaf: its folder label, the model's top guess with confidence (red = wrong), and its mixing index.

The table below uses the **validation set**, not the 14 test images. It holds 149 corn images (51 GLS + 98 NLB), grouped two ways: by folder label, and by whether the model classified them correctly. The model got 132 right and 17 wrong.

Validation corn images (149 in total)	Images	Median mixing index
By folder: GLS	51	0.02
By folder: NLB	98	0.22
By prediction: correct	132	0.12
By prediction: misclassified	17	**0.41**

- **One-directional mixing:** 54% of NLB-labelled validation leaves also have a GLS-like region (index > 0.2), against 6% of GLS leaves. That is why NLB gets called GLS and rarely the reverse.
- **Errors happen on mixed leaves:** the 17 misclassified validation images are more than three times as mixed as the 132 correct ones.
- **The 5 wrong test images are not all alike:**
  - #2 (0.50, the highest of all 163 images) and #7 (0.30) show an NLB lesion plus a separate GLS-like area: plausibly both diseases.
  - #6 (0.11) and #11 (0.18) are mostly GLS-like with a small NLB-like part: unclear.
  - #8 (0.02) is GLS-like almost everywhere: more likely a mislabel or a GLS-only leaf.
- **Some correctly classified test leaves look mixed too:** #12 (0.46), #13 (0.36) and #5 (0.28).

Conclusion and limitations

Co-infection is a plausible explanation for many of the GLS/NLB confusions, and it fits the data better than "too few images". It doesn't explain every error: some look like label noise. This is supporting evidence, not proof.

- **"GLS-like" only means similar to the GLS training images.** Other spots, such as rust, could contribute to it.

- **Small sample:** across the 14 test images the difference from validation is not significant (p = 0.09).
- **Coarse maps:** 8×8 cells of 32×32 pixels, so small spots aren't resolved individually.
- **Confirmation needs an expert:** a plant pathologist would have to examine the leaves, or the data would need multi-label annotation.

Suggested wording for the report:

> Class activation maps show that misclassified corn leaves contain spatially separate regions of Northern Leaf Blight and Gray leaf spot evidence (median mixing index 0.41 vs 0.12 for correctly classified leaves). 54% of NLB-labelled validation leaves show this pattern, which suggests co-infection that PlantVillage's single-label annotation cannot represent. Some errors, such as test image #8, show uniform GLS evidence instead, consistent with label noise.

How this was made

- **AI assistant:** analysis, code and write-up by Claude Opus 5.5 (model ID `claude-opus-5-5`, Anthropic), running in Claude Code in VS Code.
- **Training:** TensorFlow 2.15.1 on an NVIDIA RTX 3060 Laptop GPU (6 GB); EfficientNetB0 pretrained on ImageNet, fine-tuned for 4 + 6 epochs.
- **Scripts:** `train_all.py` (21-class model and test predictions), `coinfection.py` (activation maps and mixing index).